Power

ESC Modules

ESC Module 1

M1 MOTOR

ESC Module 2

M2 MOTOR

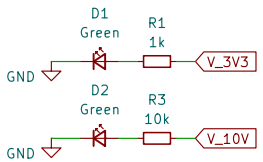
ESC Module 3

M3 MOTOR

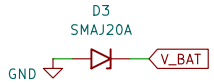
ESC Module 4

M4 MOTOR

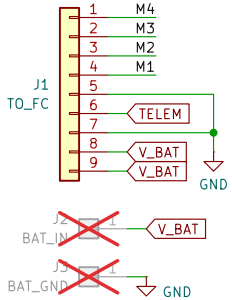
Indicators



TVS



Conns



H1 NC

H2 NC

H3 NC

H4 NC

This board placed below FC
~20A continuous per motor, ~80A total continuous
Caps on 10/5V rails: 16V+; VBAT: 30V+

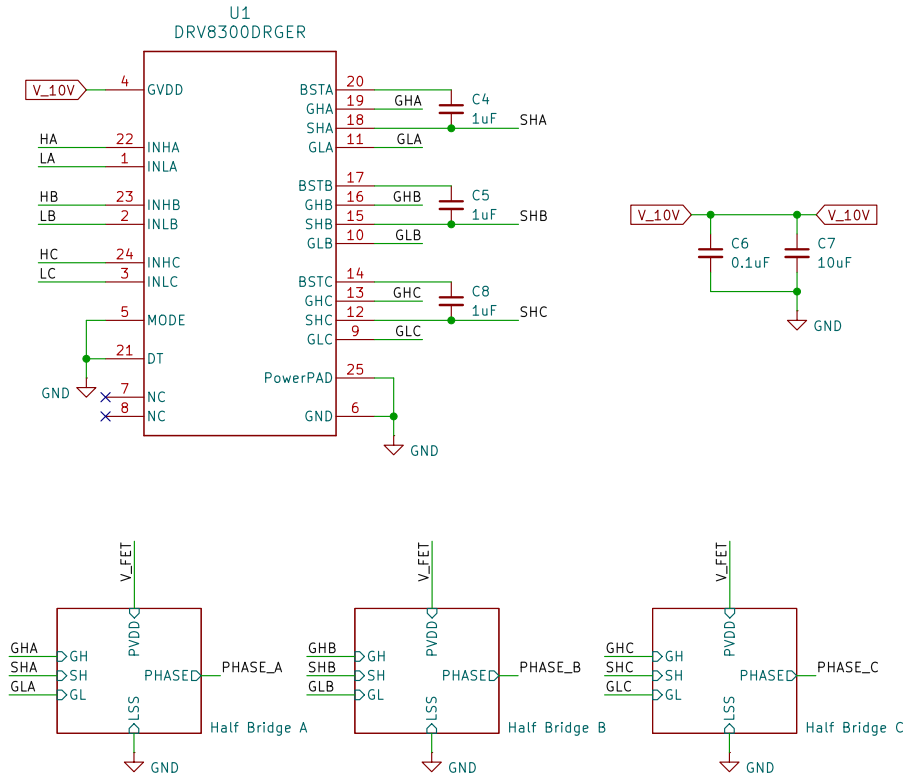
Stratus

Sheet: /
File: esc.kicad_sch

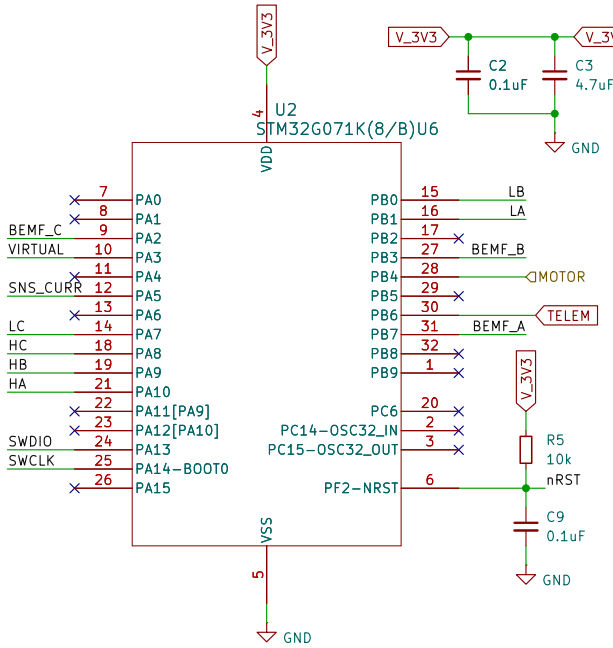
Title: 2–4S 4–in–1 BLDC ESC

Size: A4	Date: 2026–09–10	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 1/18

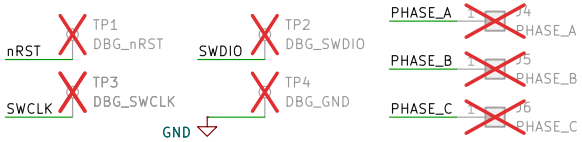
Gate driver + half bridges



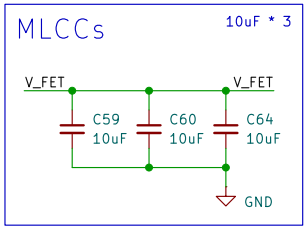
MCU



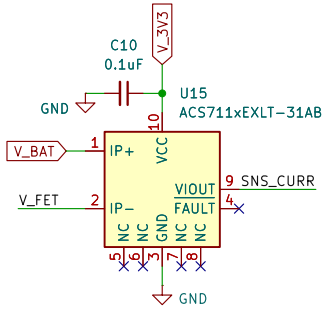
Conns



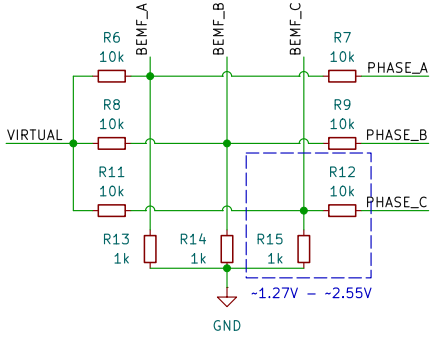
MLCCs



Current sensing



Back-EMF sensing

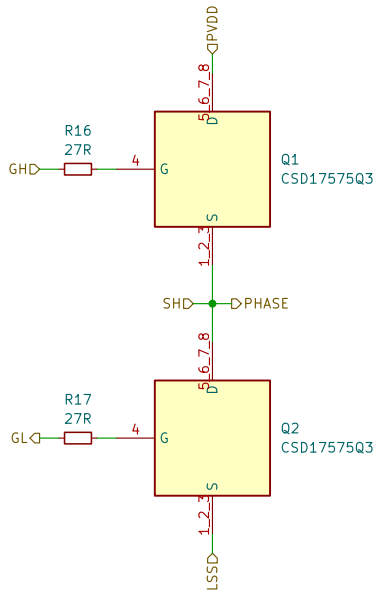


Stratus

Sheet: /ESC Module 1/
File: esc_module.kicad_sch

Title: ESC Submodule

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/18

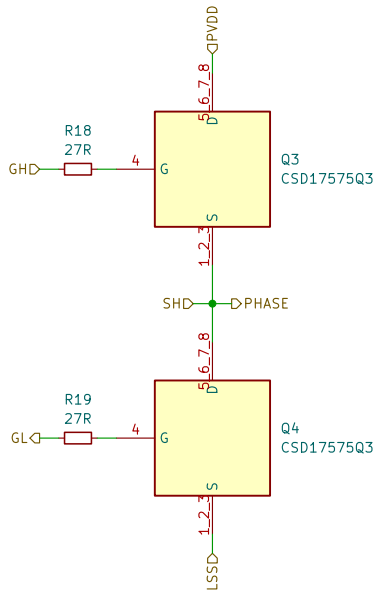


Stratus

Sheet: /ESC Module 1/Half Bridge A/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/18

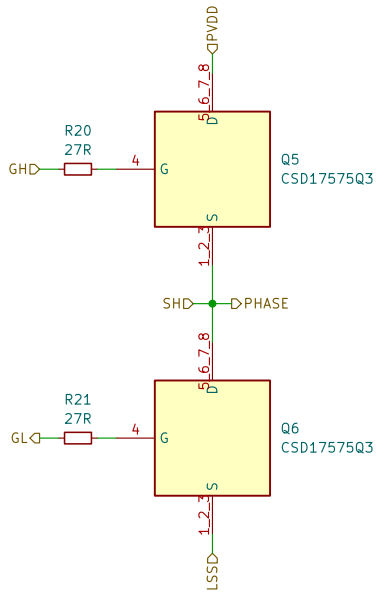


Stratus

Sheet: /ESC Module 1/Half Bridge B/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/18



Stratus

Sheet: /ESC Module 1/Half Bridge C/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/18

Gate driver + half bridges

U4
DRV8300DRGER

V_10V
GVDD
HA 22 INHA
LA 1 INLA
HB 23 INHB
LB 2 INLB
HC 24 INHC
LC 3 INLC
5 MODE
21 DT
GND
7 NC
8 NC
BSTA 20
GHA 19 GHA
SHA 18
GLA 11 GLA
C13 1uF SHA
BSTB 17
GHB 16 GHB
SHB 15
GLB 10 GLB
C14 1uF SHB
BSTC 14
GHC 13 GHC
SHC 12
GLC 9 GLC
C17 1uF SHC
PowerPAD 25
GND 6
GND

V_10V
C15 0.1uF
C16 10uF
GND

V_FET
GHA GH
SHA SH
GLA GL
PVDD
PHASED
PHASE_A
LSS
GND
Half Bridge A

V_FET
GHB GH
SHB SH
GLB GL
PVDD
PHASED
PHASE_B
LSS
GND
Half Bridge B

V_FET
GHC GH
SHC SH
GLC GL
PVDD
PHASED
PHASE_C
LSS
GND
Half Bridge C

MCU

V_3V3
U5
STM32G071K(8/B)U6
VDD
V_3V3
C11 0.1uF
C12 4.7uF
GND

BEMF_C PA0
VIRTUAL PA1
SNS_CURR PA2
PA3
PA4
PA5
PA6
LC PA7
HC PA8
HB PA9
HA PA10
PA11[PA9]
PA12[PA10]
SWDIO PA13
SWCLK PA14-BOOT0
PA15

PB0 LB
PB1 LA
PB2
PB3 BEMF_B
PB4 MOTOR
PB5
PB6 TELEM
PB7 BEMF_A
PB8
PB9
PC6
PC14-OSC32_IN
PC15-OSC32_OUT
PF2-NRST

V_3V3
R22 10k
nRST
C18 0.1uF
GND

VSS
GND

Conns

nRST TP5 DBG_nRST
SWCLK TP7 DBG_SWCLK
SWDIO TP6 DBG_SWDIO
TP8 DBG_GND
GND

PHASE_A J7 PHASE_A
PHASE_B J8 PHASE_B
PHASE_C J9 PHASE_C

MLCCs

10uF * 3

V_FET
C73 10uF
C74 10uF
C75 10uF
GND

Current sensing

C19 0.1uF
V_3V3
GND
U16
ACS711xEXLT-31AB
V_BAT 1 IP+
V_FET 2 IP-
VCC 10
VIOUT 9 SNS_CURR
FAULT 4
NC NC GND NC NC
5 6 3 7 8
GND

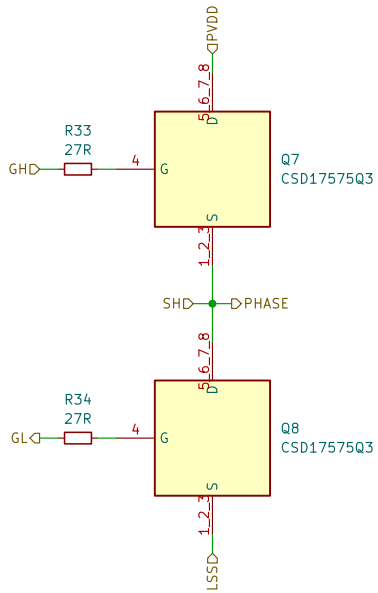
Back-EMF sensing

BEMF_A BEMF_B BEMF_C
VIRTUAL
R23 10k
R24 10k PHASE_A
R25 10k
R26 10k PHASE_B
R28 10k
R29 10k PHASE_C
R30 1k
R31 1k
R32 1k
~1.27V – ~2.55V
GND

Stratus
Sheet: /ESC Module 2/
File: esc_module.kicad_sch

Title: ESC Submodule

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/18

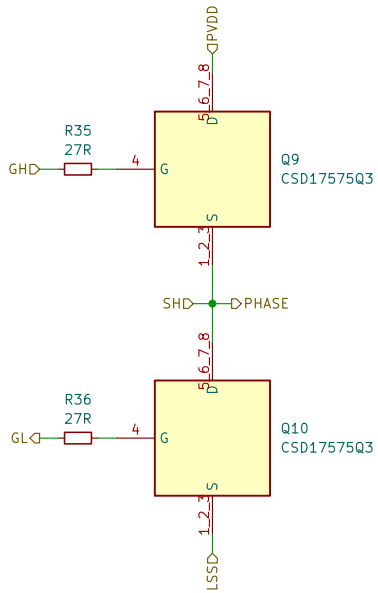


Stratus

Sheet: /ESC Module 2/Half Bridge B/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/18

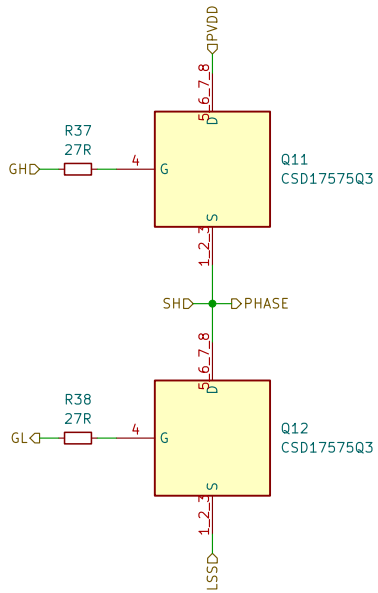


Stratus

Sheet: /ESC Module 2/Half Bridge C/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/18



Stratus

Sheet: /ESC Module 2/Half Bridge A/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 9/18

Gate driver + half bridges

U7
DRV8300DRGER

V_10V
GVDD
HA 22 INHA
LA 1 INLA
HB 23 INHB
LB 2 INLB
HC 24 INHC
LC 3 INLC
MODE 5
DT 21
NC 7
NC 8
PowerPAD 25
GND 6

BSTA 20
GHA 19 — C22 1uF — SHA
SHA 18
GLA 11
BSTB 17
GHB 16 — C23 1uF — SHB
SHB 15
GLB 10
BSTC 14
GHC 13 — C26 1uF — SHC
SHC 12
GLC 9

V_10V — C24 0.1uF — GND
V_10V — C25 10uF — GND

V_FET
GHA, SHA, GLA → PVDD, GH, SH, LSS, PHASED → PHASE_A
Half Bridge A

GHB, SHB, GLB → PHASE_B
Half Bridge B

GHC, SHC, GLC → PHASE_C
Half Bridge C

MCU

U8
STM32G071K(8/B)U6

V_3V3
VDD
VSS
GND

V_3V3 — C20 0.1uF — GND
V_3V3 — C21 4.7uF — GND

PA0 7
PA1 8
BEMF_C PA2 9
VIRTUAL PA3 10
PA4 11
SNS_CURR PA5 12
PA6 13
LC PA7 14
HC PA8 18
HB PA9 19
HA PA10 21
PA11[PA9] 22
PA12[PA10] 23
SWDIO PA13 24
SWCLK PA14-BOOT0 25
PA15 26

PB0 15 LB
PB1 16 LA
PB2 17
PB3 27 BEMF_B
PB4 28 MOTOR
PB5 29
PB6 30 TELEM
PB7 31 BEMF_A
PB8 32
PB9 1
PC6 20
PC14-OSC32_IN 2
PC15-OSC32_OUT 3
PF2-NRST 6

V_3V3 — R39 10k — nRST
C27 0.1uF — GND

Conns

nRST — TP9 DBG_nRST
SWDIO — TP10 DBG_SWDIO
SWCLK — TP11 DBG_SWCLK
TP12 DBG_GND — GND

PHASE_A — J10 PHASE_A
PHASE_B — J11 PHASE_B
PHASE_C — J12 PHASE_C

MLCCs

10uF * 3

V_FET — C76 10uF — C77 10uF — C78 10uF — V_FET
GND

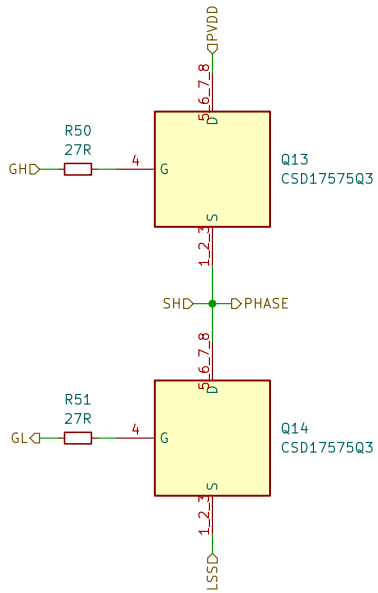
Current sensing

C28 0.1uF
V_3V3
GND
U17
ACS711xEXLT-31AB
V_BAT → IP+ 1
VCC 10
V_FET → IP− 2
VIOUT 9 SNS_CURR
FAULT 4
NC NC GND NC NC
GND

Back-EMF sensing

BEMF_A, BEMF_B, BEMF_C
VIRTUAL
R40 10k
R41 10k → PHASE_A
R42 10k
R43 10k → PHASE_B
R45 10k
R46 10k → PHASE_C
R47 1k
R48 1k
R49 1k
~1.27V – ~2.55V
GND

Stratus	
Sheet: /ESC Module 3/	
File: esc_module.kicad_sch	
Title: ESC Submodule	
Size: A4	Date: 2026-07-06
KiCad E.D.A. 10.0.6	Id: 10/18
	Rev: A

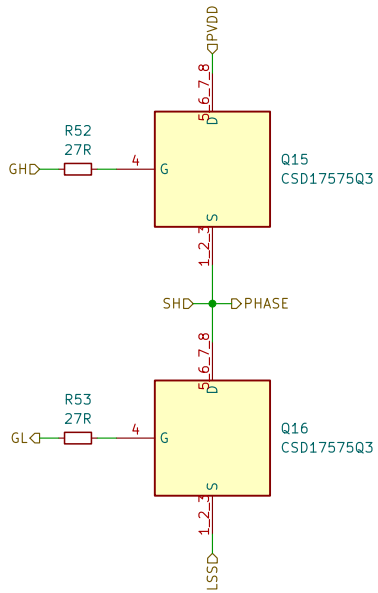


Stratus

Sheet: /ESC Module 3/Half Bridge B/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 11/18

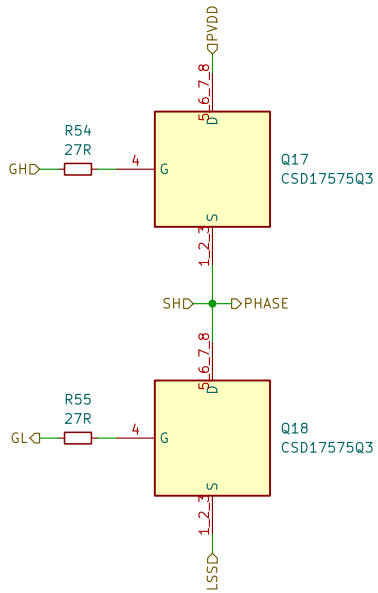


Stratus

Sheet: /ESC Module 3/Half Bridge C/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 12/18



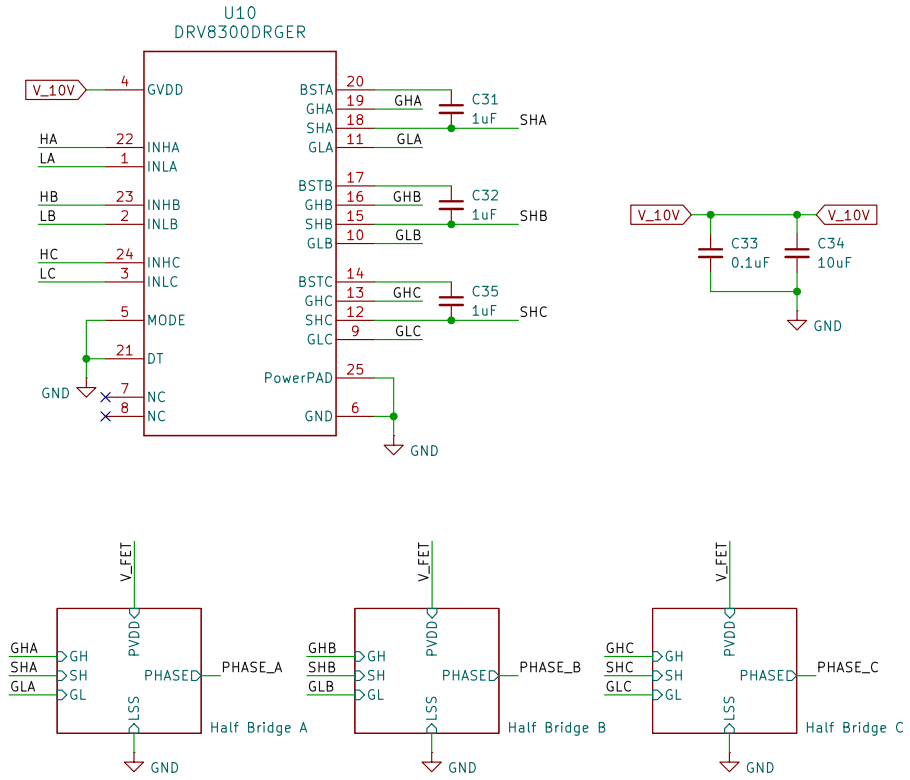
Stratus

Sheet: /ESC Module 3/Half Bridge A/
File: half_bridge.kicad_sch

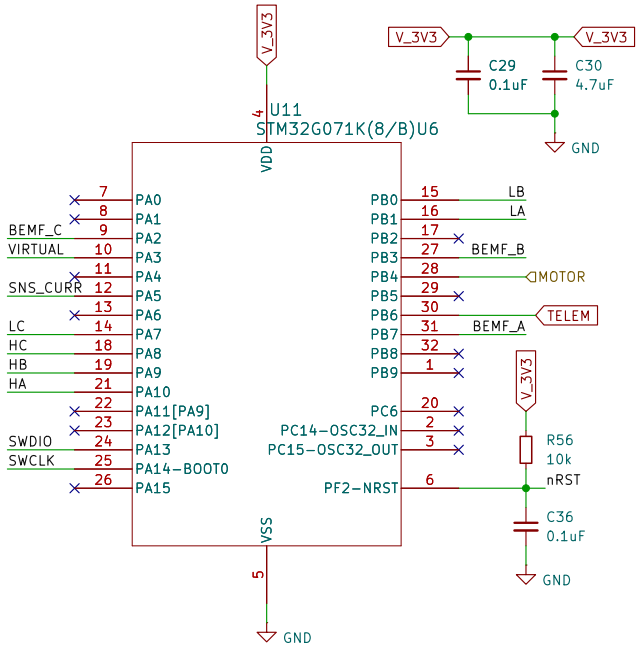
Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 13/18

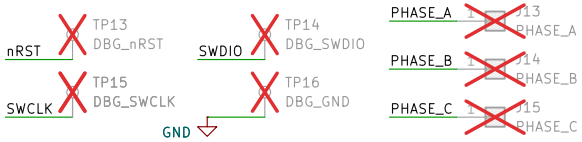
Gate driver + half bridges



MCU

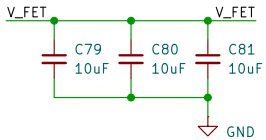


Conns

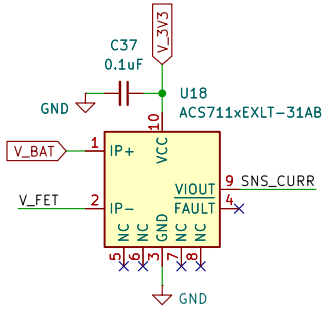


MLCCs

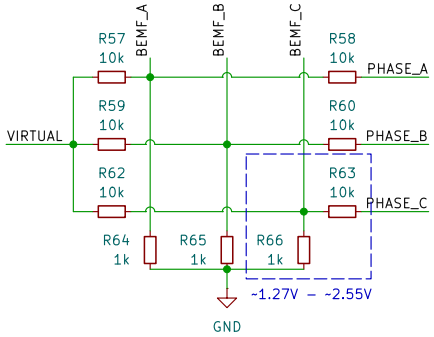
10uF * 3



Current sensing



Back-EMF sensing

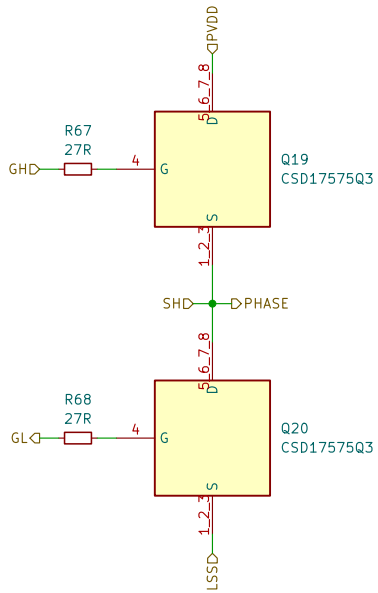


Stratus

Sheet: /ESC Module 4/
File: esc_module.kicad_sch

Title: ESC Submodule

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 14/18

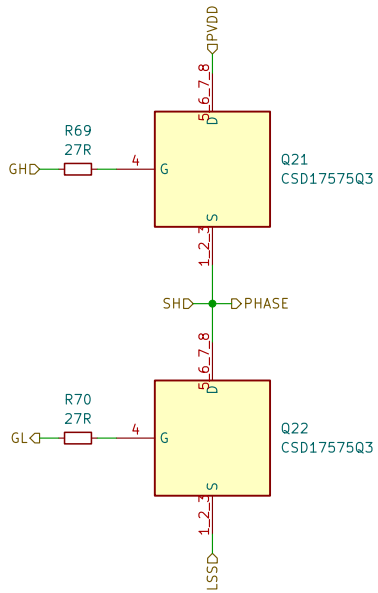


Stratus

Sheet: /ESC Module 4/Half Bridge B/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 15/18

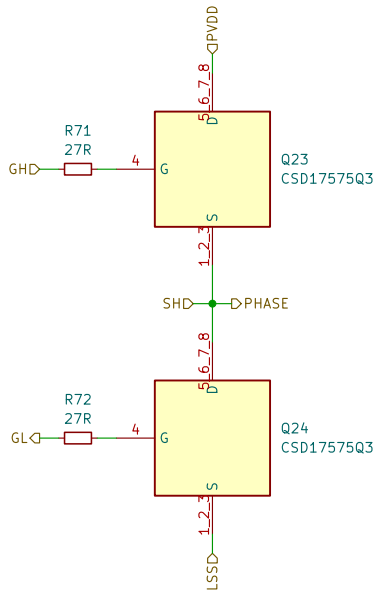


Stratus

Sheet: /ESC Module 4/Half Bridge C/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 16/18



Stratus

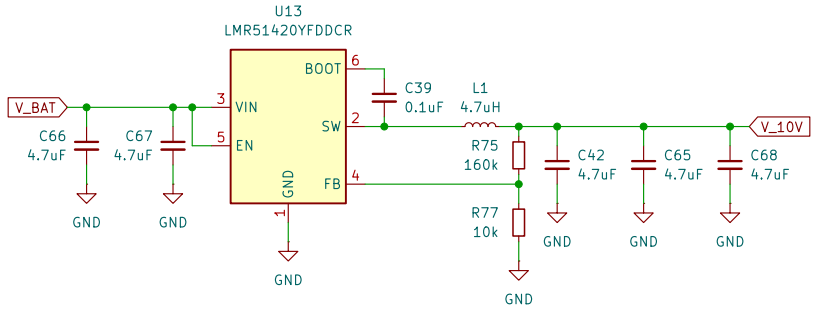
Sheet: /ESC Module 4/Half Bridge A/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 17/18

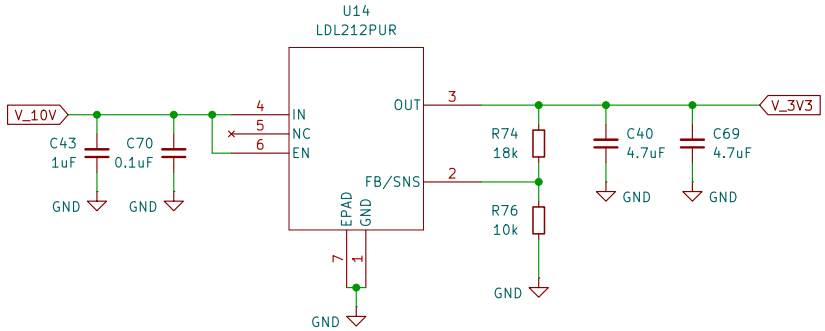
Buck: V_BAT -> 10V

to DRV8300 and LDO



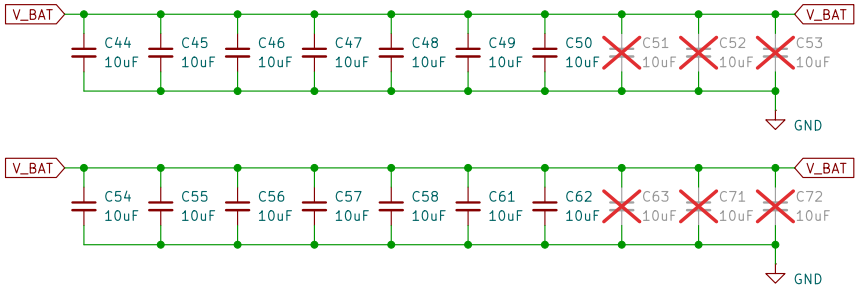
LDO: 10V -> 3V3

to STM32s



Capacitor bank

10uF * 12



Stratus		
Sheet: /Power/		
File: power.kicad_sch		
Title: Power Stages		
Size: A4	Date: 2026-09-10	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 18/18